

To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main() {
    char path[100];
    char oldpath[100];

    |   getcwd(oldpath, sizeof(oldpath));

    printf("Current directory: %s\n", oldpath);

    printf("Enter directory to change to: ");
    scanf("%s", path);

    if (chdir(path) == 0) {
        printf("Directory changed successfully.\n");

        getcwd(path, sizeof(path));
        printf("New directory: %s\n", path);

        printf("Previous directory: %s\n", oldpath);
    }
    else {
        perror("Error changing directory");
    }

    return 0;
}
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 4$ ./task3
Current directory: /home/geya/OS/Skill/Week 4
Enter directory to change to: /Week 3
Error changing directory: No such file or directory
geya@LenovoL480:~/OS/Skill/Week 4$ ./task3
Current directory: /home/geya/OS/Skill/Week 4
Enter directory to change to: /tmp
Directory changed successfully.
New directory: /tmp
Previous directory: /home/geya/OS/Skill/Week 4
geya@LenovoL480:~/OS/Skill/Week 4$ |
```